

Asher Siddiqui

+923302463301 | ashersiddique127@gmail.com | [linkedIn](#) | [github.com/pluuto19](#)

EDUCATION

FAST-NUCES Bachelor of Software Engineering

Expected Graduation: June 2025

Key Courses: Computer Networks, Database Systems & Design, Design & Analysis of Algorithms, Operating Systems, Data Structures, Web Engineering, Object-Oriented Programming, Software Design & Architecture, Software Quality Assurance

SKILLS

Programming Languages: Go-Lang, Javascript (NodeJS & Vanilla), Bash, Java, C-Lang, Oracle Database, MongoDB, HTML/CSS

Libraries & Tools: Git, Docker, Linux, WSL, ExpressJS, JavaFX

WORK EXPERIENCE

FAST-NUCES

Teaching Assistant - Operating Systems

- Demonstrated strong organization, communication, and time management skills by efficiently working with instructors to manage and maintain the grades of 90 students.
- Programmed an assignment evaluation tool, producing significant time savings and highlighting problem-solving skills.

PROJECTS

Load Balancer [Go Lang, ExpressJS, Docker, Git, HTML]

- Designed and implemented a high-throughput HTTP reverse proxy, achieving efficient traffic distribution as demonstrated by the integration of round-robin and least connection algorithms.
- Enhanced application scalability and reduced computation overhead, as demonstrated by a throughput of 10000 requests per 5 seconds, by utilizing GOROUTINES and buffered channels.
- Boosted the load balancer's reliability by implementing health checks to ensure the backend servers' availability.

Project Management System [ExpressJS, MySQL, Handlebars, Git]

- Developed a robust webserver architecture for a project management system using ExpressJS and MySQL, utilizing the microservice architecture with three distinct services.
- Facilitated seamless interaction and data flow within the application by orchestrating communication between services through the integration of the controller with backend APIs.
- Secured the system's services by employing API keys and JWT authentication with the REST APIs and the controller, respectively.

Visualizing Convex Hulls [Java, JavaFX, Git]

- Demonstrated practical skills by translating research papers into applications using JavaFX.
- Conducted an analysis of computational complexity by implementing and executing various convex hull algorithms.

More Projects [Python, BASH, ExpressJS, MySQL, Java, Java Swing, Git, HTML]

- **Pyonic Reader:** Developed a Python CLI tool to parse PDFs and bold the initial letters of every word to improve the reading focus of people with ADHD, based on research exploring the effectiveness of typographical cues for improving readability.
- **Assess Buddy:** Coded a bash tool to streamline the extraction, compilation, and execution of compressed folders and source code, facilitating the evaluation of student-submitted assignments.
- **Rent a Car & Autoshop:** Developed an ExpressJS-based app with a normalized MySQL database for efficient CRUD operations.
- **A Night in the Woods:** Designed and implemented a 2D platformer game made entirely in Java Swing, implementing multithreading to support background audio.

LEADERSHIP EXPERIENCE

Microsoft Learn Students Ambassador

Technical Coordinator

- Coordinated technical activities for a workshop with approximately 50 participants. Curated and managed workshop materials for the audience.
- Actively assisted the event management team during the workshop and ensured smooth execution of the workshop.

VOLUNTEER EXPERIENCE

Destination Imagination Pakistan 2024: Acted as a team guide and mentored K-12 students at Destination Imagination Pakistan 2024, encouraging creativity and problem-solving skills. Contributed to a positive team experience, resulting in strengthened leadership and mentorship abilities.

AWARDS & HONORS

Dean's List of Honors: Fall 2023 (SGPA: 3.73/4.00), Spring 2024 (SGPA: 3.80/4.00)